Optical Test Structures for Waveguide Sidewall Angle: Why Strips Are Blind and How Rib Width Sets Recover the Angle

Unmol Sharma and David [LAST NAME]

***Abstract*—The sidewall angle of etched silicon waveguides changes the effective index of every guided mode, yet the optical test structures used to monitor fabrication do not report it. We analyze which test structures can determine the angle, using full-vector finite-element mode solutions and the Cramér–Rao lower bound (CRLB) of a joint fit of ring and interferometer readouts. For an oxide-clad strip the mode is mirror-symmetric about mid-height, so to first order a wall tilt θ about the bottom edge is exactly a width change of $t\tan\theta$, $3.84\,\mathrm{nm}$ per degree for a $220\,\mathrm{nm}$ film: sets of strips carry no angle information at any size. A partially etched rib breaks the symmetry, with an angle/width ratio of 0.75–0.87 instead of 1, but a rib made by a separate etch brings its own depth and angle, and one rib width then leaves the strip angle unidentifiable. Two widths of the same rib resolve it. With strip rings at five widths and ribs of $300\,\mathrm{nm}$ and $500\,\mathrm{nm}$ on a $50\,\mathrm{nm}$ slab, the CRLB on the strip sidewall angle is 0.39° for a $1\,\mathrm{pm}$ resonance-wavelength precision, with etch depth, rib angle and both material indices unknown; the best eight-structure set reaches about 0.12–0.14°, with a $15\,\%$ uncertainty from derivative accuracy. Nonlinear fits of 2000 simulated noisy measurement sets reproduce these bounds to within $5\,\%$. Treating a 1° tilt as a width error mispredicts a rib ring resonance by $711\,\mathrm{pm}$. The results are from simulations only.**

***Index Terms*—Silicon photonics, sidewall angle, process monitoring, test structures, Cramér–Rao bound, ring resonators.**

I. Introduction

Silicon photonic circuits are designed around nominal waveguide cross-sections, and their behavior follows the fabricated ones. Width and thickness variations across a wafer shift ring resonances and interferometer phases, and models of layout-dependent variability are now used to predict circuit yield [1]. The sidewall angle is the least monitored of these parameters. Plasma etching leaves walls that lean away from vertical, the lean varies with etch chemistry, pattern density and position on the wafer

cite: measured sidewall-angle variation in silicon photonic etch processes

, and it moves the effective index of every mode that touches the wall. Circuits that mix fully etched strips with partially etched ribs see two different angles from two different etch steps, so a component designed for one cross-section can sit off resonance when either angle drifts.

U. Sharma is with Vanderbilt University, Nashville, TN, USA (e-mail:

e-mail

).

D. [LAST NAME] is with the SOL Lab, Vanderbilt University, Nashville, TN, USA.

affiliation details, corresponding author, manuscript dates

Dimensional metrology measures the angle directly but sparsely. Cross-sectional electron microscopy requires cutting the sample and is destructive [2]; atomic force microscopy needs specialized tips and scans to reach a wall

cite: 3D/CD-AFM sidewall metrology on Si waveguides

; optical scatterometry is fast and non-destructive but infers the profile through a parametrized model whose parameters can be correlated and whose inverse problem need not have a unique solution [2], [3]. All of these measure shape at a few sites. What a circuit designer needs is the optical consequence of the shape across the wafer, which suggests measuring the angle with the same ring and interferometer test structures that already monitor width and thickness. Such optical extraction methods recover width and thickness from measured effective and group indices [4], [5], but they hold the sidewall angle at an assumed value

verify in the full text of Xing et al. 2018 that the extraction fixes the sidewall angle (search snippet: 85° from a foundry handbook)

.

The reason the angle is usually fixed is not a lack of data but a property of the structures. This paper shows that for oxide-clad strip waveguides the angle is not merely hard to separate from the width: to first order it cannot be separated at all, for any number of strips, polarizations or readouts. We then ask which structures restore the missing information, how much prior knowledge of the process they need, and what precision they reach under a measurement noise taken from published ring-resonator metrology [6]. The answers come from full-vector finite-element mode solutions and from the Cramér–Rao lower bound (CRLB) of a joint fit, checked by nonlinear fits of simulated noisy data. No devices were fabricated; every number below is a simulation result or an input taken from the cited literature.

The contributions are:

- a first-order proof, confirmed numerically, that oxide-clad strips are exactly blind to the sidewall angle, with the angle/width ratio $R = (y_c - y_{\mathrm{pivot}})/(h/2)$ set by the wall-perturbation centroid of the mode;
- an identifiability analysis showing that one rib made by a separate etch cannot determine the strip angle, and that prior knowledge of the etch depth does not change this;
- test-structure sets built from several widths of the same rib that reach 0.39° (two rib widths) and 0.12–0.14° (best set of eight) at a $1\,\mathrm{pm}$ wavelength precision, with the bound confirmed by Monte Carlo fits;

- a methods comparison showing that finite-difference derivatives from a grid-based solver were not accurate enough for this analysis, and that small derivative errors in any solver create false angle information unless the strip symmetry is imposed.

II. Theory

A. Measurement model and Cramér–Rao bound

Each test structure is a waveguide of bottom width w in a silicon film of thickness t, read out by a ring resonator and a Mach–Zehnder interferometer (MZI). The unknown parameters form a vector $\mathbf{p}$ that always contains the strip width offset, the film thickness and the strip sidewall angle θ, measured from vertical with the top narrower. A ring gives two observables, the effective index n_{eff} through the resonance position and the group index n_{g} through the free spectral range; an MZI gives one phase. Reading n_{eff} from a resonance position assumes that the resonance order and the ring length are known, so that a wavelength shift maps onto an index change; we make this assumption throughout. Stacking all observables into $\mathbf{y} = \mathbf{f}(\mathbf{p}) + \boldsymbol{\epsilon}$ with independent Gaussian noise of standard deviations σ_k, the Fisher information matrix is

$$\mathbf{F} = \mathbf{J}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \mathbf{J} + \boldsymbol{\Lambda}, \qquad J_{kj} = \frac{\partial f_k}{\partial p_j}, \tag{1}$$

where $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_k^2)$ and $\boldsymbol{\Lambda}$ is diagonal in the parameters that have a Gaussian prior, with entries $1/\sigma_{p,j}^2$ (a prior on a difference of two parameters adds the corresponding rank-one term). The CRLB on the angle is [7]

$$\sigma_\theta \geq \sqrt{[\mathbf{F}^{-1}]_{\theta\theta}}. \tag{2}$$

When $\mathbf{F}$ is singular along a direction that has a component on θ, no unbiased estimator has finite variance and we call the angle unidentifiable ($\sigma_\theta = \infty$). Numerically, $\mathbf{J}$ is column-normalized and singular values below a fixed small fraction of the largest are treated as zero.

The noise is set by the precision σ_λ with which a resonance wavelength can be located. Janz *et al.* report that wavelength changes of $1\,\mathrm{pm}$ or less are routinely measured with calibrated tunable lasers, and that a temperature-stabilized ring reached a $0.1\,\mathrm{pm}$ uncertainty set by the laser repeatability [6]. We take $1\,\mathrm{pm}$ as the primary case and $0.1\,\mathrm{pm}$ as the best case. The conversion to the three observables is ours. A resonance shift obeys $\delta\lambda/\lambda = \delta n_{\mathrm{eff}}/n_{\mathrm{g}}$, so $\sigma_{n_{\mathrm{eff}}} = n_{\mathrm{g}}\sigma_\lambda/\lambda$; the group index follows from one free spectral range, $n_{\mathrm{g}} = \lambda^2/(L\,\mathrm{FSR})$, measured with two resonances, so $\sigma_{n_{\mathrm{g}}} = \sqrt{2}\,\sigma_\lambda n_{\mathrm{g}}^2 L/\lambda^2$; and an MZI fringe read as precisely as a ring resonance gives $\sigma_\phi = 2\pi n_{\mathrm{g}} L \sigma_\lambda/\lambda^2$. With $\lambda = 1.55\,\mu\mathrm{m}$, a ring and MZI length $L = 100\,\mu\mathrm{m}$ and $n_{\mathrm{g}} = 4.26$, the largest group index among the structures used, $\sigma_\lambda = 1\,\mathrm{pm}$ gives $\sigma_{n_{\mathrm{eff}}} = 2.75 \times 10^{-6}$, $\sigma_{n_{\mathrm{g}}} = 1.07 \times 10^{-3}$ and $\sigma_\phi = 1.11 \times 10^{-3}\,\mathrm{rad}$. These three values are derived, not measured; the MZI value in particular assumes that a broad fringe is located as well as a narrow resonance. At $0.1\,\mathrm{pm}$ the ring must also be held to about $1.25\,\mathrm{mK}$, since its resonance moves by roughly $80\,\mathrm{pm/K}$ [6].

B. Unknown parameters and priors

We add process unknowns in stages. Stage B adds the etch depth d of a partially etched rib, with the rib angle taken equal to θ. Stage C gives the rib its own angle θ_r, as when the partial etch is a separate step. Stage D adds the silicon and oxide indices n_{Si} and n_{ox}, and stage E the temperature T, which enters through the thermo-optic coefficients of silicon [8] and silica [9]. Unless stated, results are for stage D. "Realistic" priors are $\sigma_d = 2\,\mathrm{nm}$, $\sigma_{n_{\mathrm{Si}}} = 10^{-4}$, $\sigma_{n_{\mathrm{ox}}} = 10^{-3}$ and $\sigma_T = 1\,\mathrm{K}$. These are assumptions; the silicon value is close to the 8.67×10^{-5} absolute index uncertainty measured for bulk silicon at $1.5\,\mu\mathrm{m}$ and $295\,\mathrm{K}$ [8], while device-layer doping, stress and deposited cladding oxides are not covered by any source we could verify.

C. Why a strip cannot see its sidewall angle

Tilting a wall by θ about its bottom edge at height y_{pivot} moves the wall inward by $\delta x(y) = (y - y_{\mathrm{pivot}})\tan\theta$. To first order, the change of n_{eff} caused by moving a dielectric boundary is a weighted integral of the displacement over the boundary [10],

$$\delta n_{\mathrm{eff}} \propto -\int_{\mathrm{wall}} W(y)\,\delta x(y)\,dy, \tag{3}$$

with $W(y) = \Delta\varepsilon\,|\mathbf{E}_\parallel|^2 + \Delta\varepsilon\,(\varepsilon_1/\varepsilon_2)\,|E_\perp|^2$ evaluated on the core side, $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$, and ε_1, ε_2 the core and cladding permittivities. A uniform width increase δw moves both walls outward by $\delta w/2$. Comparing the two perturbations for both walls of a symmetric waveguide gives

$$\frac{\partial n_{\mathrm{eff}}/\partial\theta}{\partial n_{\mathrm{eff}}/\partial w} = -h\,R\,\tan(1^\circ)\ \text{per degree}, \tag{4}$$

$$R = \frac{y_c - y_{\mathrm{pivot}}}{h/2}, \tag{5}$$

where h is the height of the tilted wall and $y_c = \int yW\,dy / \int W\,dy$ is the centroid of the wall weight. We write (4) in the form of a central difference over $\pm 1^\circ$, which is how the derivatives are computed; the exact derivative has $\pi/180$ in place of $\tan(1^\circ)$. A tilt therefore acts exactly like a width change of $-2(y_c - y_{\mathrm{pivot}})\tan\theta$.

For a strip with oxide both below and above, the structure and its mode are mirror-symmetric about mid-height, so $y_c = t/2$ and $R = 1$ exactly. A 1° tilt is then indistinguishable from a width change of $t\tan(1^\circ) = 3.8401\,\mathrm{nm}$ for $t = 220\,\mathrm{nm}$. The relation holds at every wavelength, so it holds for n_{g} as well as for n_{eff}, for TE and TM, and for every strip width. The angle column of $\mathbf{J}$ is then a fixed multiple of the width column for every strip observable, $\mathbf{F}$ is singular, and no set of strips determines θ. The rotation of the wall about mid-height that remains after removing the uniform shift is odd under the mirror operation and enters only at second order.

A rib breaks the mirror symmetry: the wall is the upper part of the cross-section, its pivot is the slab top, and the slab pulls the mode down toward the wall's lower corner. Then $y_c < y_{\mathrm{pivot}} + h/2$ and $R < 1$, and the angle column of a rib is no longer parallel to its width column. Whether this new direction survives in a joint fit depends on how many unknowns the rib brings, which we count in Section III.

III. Test Structures

All structures share a $220\,\mathrm{nm}$ silicon film with oxide below and an oxide cladding above. Strips are fully etched, with bottom widths of 400, 450, 500, 550 and $600\,\mathrm{nm}$ in both TE and TM. Ribs are TE only, with a slab of thickness s left by a partial etch of depth $d = t - s$. At $s = 70$ and $90\,\mathrm{nm}$ we simulated rib widths of 300, 400, 500 and $600\,\mathrm{nm}$; at $s = 30$, 40, 50, 110, 130 and $150\,\mathrm{nm}$ we simulated widths of 300 and $500\,\mathrm{nm}$, which is enough to compare slabs for a two-width set. Each structure is read out by one ring and one MZI.

The MZI turns out to be redundant with a ring of the same waveguide. Its phase is proportional to n_{eff}, so its row in $\mathbf{J}$ is parallel to the ring's n_{eff} row and only lowers the effective n_{eff} noise. For the two sets discussed in Section V, the CRLB with ring and MZI equals the CRLB of the ring alone with the combined n_{eff} noise to better than one part in 10^{12}; removing the MZI raises the bound of the best eight-structure set from 0.139° to 0.190° and that of the two-width set from 0.389° to 0.393°. An MZI is therefore worth adding only as a cheaper way to average down the n_{eff} noise.

Identifiability follows from a simple count. Strips share w, t and θ, and every strip adds two observables but no unknowns; yet their angle and width columns are parallel (Section II), so they constrain only one combination of the two. A rib from a separate etch adds two observables (n_{eff}, n_{g}) and two unknowns (d, θ_r). Its observables are used up determining its own etch, and nothing is left over for θ. A second width of the same rib adds two observables and no unknowns, and the two widths sense d and θ_r with different weights, which frees the rib's width information to break the strip degeneracy. Stage B, where the rib shares the strip angle, is the exception: there one rib adds only one unknown.

IV. Simulation Methods

A. Mode solver and geometry

Modes were computed with femwell [11], a full-vector finite-element mode solver built on scikit-fem [12], on conforming triangular meshes generated by Gmsh [13]. The mesh follows every material boundary, so a perturbed wall is represented exactly rather than by a staircase. The production mesh ("r12") has a $12\,\mathrm{nm}$ element size in the core. The domain spans $|x| \leq 1.4\,\mu\mathrm{m}$ and $|y| \leq 1.2\,\mu\mathrm{m}$, with silicon $n_{\mathrm{Si}} = 3.48$ and oxide $n_{\mathrm{ox}} = 1.44$ for both the buried oxide and the cladding, without material dispersion, at $\lambda = 1.55\,\mu\mathrm{m}$. These indices are rounded assumptions; the measured bulk values at $1.5\,\mu\mathrm{m}$ and $295\,\mathrm{K}$ are 3.48314 for silicon [8] and 1.44502 for fused silica [9]. The width w is the bottom width of the wall; the angle tilts the wall about its bottom edge, which for a rib is the slab top. Walls are ideal trapezoids. For each geometry we solve for two modes near a guess and keep the highest-index mode of the requested polarization, falling back to a six-mode search without a guess if that mode is more than $3\,\%$ from the guess. The group index is $n_{\mathrm{g}} = n_{\mathrm{eff}} - \lambda\,\partial n_{\mathrm{eff}}/\partial\lambda$ from a central difference over $\pm 5\,\mathrm{nm}$ on the same mesh.

B. Derivatives and their checks

The Jacobian in (1) is built from central differences, each perturbed geometry being remeshed, with steps h of $5\,\mathrm{nm}$ in width, $2\,\mathrm{nm}$ in thickness and slab, 1° in angle and 0.002 in each index. Every derivative was also computed with steps $h/2$ and $2h$. Of the 428 derivatives, 422 agree to within $2\,\%$ over the three steps, 3 more are near zero and agree in absolute terms, and 3 exceed the limit slightly, the largest by $2.25\,\%$; all three are n_{g} derivatives. Bounds are reported at step h with the range over $h/2$, h and $2h$.

Refining the mesh from r12 to r8 ($8\,\mathrm{nm}$ core elements) changes no derivative of the strip and rib TE $500\,\mathrm{nm}$ structures by more than $0.31\,\%$. Because remeshing every perturbed geometry adds solve-to-solve noise, we also recomputed the geometric derivatives of the $300\,\mathrm{nm}$ ribs, the most weakly confined structures, on a fixed mesh: the nominal mesh is reused and its nodes are moved onto each perturbed geometry by a continuous piecewise-linear map that keeps every material boundary on its line. The fixed-mesh derivatives differ from the remeshed ones by at most $0.13\,\%$ on the $50\,\mathrm{nm}$ slab and $0.78\,\%$ on the $90\,\mathrm{nm}$ slab. Enlarging the computational domain by a factor of 1.5 in each direction, to $|x| \leq 2.1\,\mu\mathrm{m}$ and $|y| \leq 1.8\,\mu\mathrm{m}$, changes n_{eff} of the seven structures of the two-width set by at most 2.04×10^{-6} and n_{g} by at most 1.32×10^{-5} in relative terms, and their geometric derivatives by at most $0.35\,\%$. Because the two-width bound responds to derivative errors of this size, it moves by $5.2\,\%$, to 0.409°, when the large-domain geometric derivatives replace the standard ones (the index derivatives are kept from the standard domain).

The mode indices agree with an earlier finite-difference solution of the same structures (EMpy [14], with cell-exact averaging): within $0.79\,\%$ in n_{eff} for the ten strips and $0.15\,\%$ for the five ribs. For the rib of $500\,\mathrm{nm}$ width on a $90\,\mathrm{nm}$ slab, femwell gives $n_{\mathrm{eff}} = 2.5707$, $1.2\,\%$ below the effective-index-method value of 2.602 reported for the same geometry and indices [15]; a full-vector solution is expected to lie below the effective-index estimate.

C. Why the strip symmetry is imposed

Agreement in n_{eff} does not imply agreement in derivatives, and the analysis depends on derivatives. Across 30 structures, the EMpy derivatives differ from the femwell ones by a median of 4.7–$14\,\%$ depending on the derivative, and in the same $2\,\%$ step-size test 222 EMpy derivatives failed (373 passed and 53 were near zero). We therefore use femwell throughout. Even femwell's derivatives are not exact: the femwell strip ratios $(\partial n_{\mathrm{eff}}/\partial\theta)/(\partial n_{\mathrm{eff}}/\partial w)$ scatter by up to $0.84\,\%$ around the exact value. That scatter alone is enough to create apparent angle information in strips. In an earlier pass of this analysis, with placeholder noise values, the best set of strips alone gave a finite 0.6135° from the raw femwell derivatives and an infinite bound once the exact relation was imposed. Since the relation is exact for oxide-clad strips (Section II), every result below replaces each strip's angle column by $-t\tan(1^\circ)$ times its width column.

TABLE I
STRIP SIDEWALL-ANGLE CRLB σ_θ FOR THE MAIN TEST-STRUCTURE SETS (STAGE D, REALISTIC PRIORS). RANGES ARE OVER THE DERIVATIVE STEPS $h/2$, h, $2h$; "MC" IS THE STANDARD DEVIATION OF 2000 NONLINEAR FITS AT 1 pm.

Set	σ_θ at 1 pm (°) h	range	0.1 pm (°)	MC (°)
Strips only, any number	∞	–	∞	–
Strips + one rib (70 nm slab)	25.19	–	–	–
Strips + rib 300, 500 (50 nm)	0.3891	0.3891–0.3997	0.04077	0.3976
Best set of 8 (90 nm)	0.139	0.1007–0.1473	0.0139	0.1323

D. Monte Carlo fits

To test whether the linearized bound is reached by an actual estimator, we fitted simulated noisy data with the full nonlinear model. Because thousands of fits need fast forward evaluations, each structure's n_{eff} and n_{g} were represented by a polynomial in its own inputs (width, thickness, angle, slab, n_{Si}, n_{ox} for ribs; mid-height width, thickness and indices for strips, which encodes the symmetry), fitted to femwell solves on a Latin hypercube in a box of ± 6 marginal CRLB standard deviations around nominal, with the polynomial degree chosen by leave-one-out error. On held-out solves the strip surrogates are within $0.33\,\sigma_\phi$ in n_{eff} (with σ_ϕ expressed as an n_{eff} noise) and the 400–600 nm ribs within 1.13–$2.24\,\sigma_\phi$. The 300 nm ribs miss by $7.23\,\sigma_\phi$ (90 nm slab) and $27\,\sigma_\phi$ (50 nm slab). For these weakly confined ribs the misses are femwell's remeshing scatter between unrelated geometries rather than lack of fit, and no smooth model can follow them; the same scatter cancels within one finite-difference stencil, as the fixed-mesh check shows. Data and fits both use the surrogate, so the Monte Carlo is self-consistent. Each case uses 2000 noise realizations at $\sigma_\lambda = 1$ pm, with prior centers drawn from the priors, and bounded nonlinear least squares from five starts (nominal and four random points in the box). A fit is flagged when it ends at the box edge, and as ambiguous when another start reaches a χ^2 within 1 of the best with a different angle.

V. RESULTS

Unless stated otherwise, bounds are for stage D (width, thickness, strip angle, etch depth, rib angle, n_{Si} and n_{ox} unknown) with realistic priors, $\sigma_\lambda = 1$ pm, at derivative step h, and the bracket gives the range over $h/2$, h and $2h$. Table I collects the main sets.

A. Strips are blind to the angle

Fig. 1 shows the mechanism. For the strip the wall weight $W(y)$ is symmetric about mid-height and its centroid sits at $y_c = t/2$; for the rib the weight is concentrated near the slab corner and the centroid lies below the middle of the wall. The femwell derivatives follow (4). Over all ten strips, in TE and TM and for both n_{eff} and n_{g}, the ratio $(\partial n/\partial\theta)/(\partial n/\partial w)$ lies between 3.8396 nm and 3.8725 nm per degree, against $t\tan(1°) = 3.8401$ nm; the largest deviation is 0.84 %. The closed-form R computed from the mode-field centroid matches the femwell ratio within 0.79 % for the strips, a median of 1.25 % for the ribs, and at worst 5.4 %. For ribs, R lies between 0.831 and 0.873 on the 50 nm slab and between 0.750 and 0.799 on the 90 nm slab, and it falls as the slab thickens (Fig. 2). With the symmetry imposed, the best set of strips of any size from one to eight has an infinite bound: strips alone cannot determine the angle.

B. One rib against unknown etch parameters

Fig. 3 follows a rib on a 70 nm slab through the stages. If the rib shares the strip angle (stage B) and only its depth is unknown, one rib width already gives 0.357° and two widths give 0.00365°. Giving the rib its own angle (stage C) changes the picture. The single rib then leaves the strip angle unidentifiable, as the count in Section III predicts, while the two-width set loses only part of its precision and gives 0.448° without priors. Adding n_{Si}, n_{ox} (stage D) and temperature (stage E) moves the two-width bound to 0.481° without priors and to 0.452° and 0.455° with realistic priors.

C. Priors do not rescue a single rib

With a single rib whose etch angle is independent of the strip's (stage C, 70 nm slab, 1 pm), the strip angle is unidentifiable without priors. Prior knowledge improves this only partially: realistic priors give $\sigma_\theta = 25.18°$, knowing the etch depth to 0.01 nm (far beyond what a separate thickness measurement provides) gives 5.03°, and tying the rib and strip angles to within 1° gives 1.25°. Two widths of the same rib reach 0.448° at the same stage without any of these assumptions. In stage D with realistic priors, the single rib gives between 24.4° and 45.4° depending on the slab; these values are set by the priors, not by the data.

D. Multi-width rib sets

The two-width set, strip TE rings at the five widths plus ribs of 300 nm and 500 nm on a 50 nm slab, gives $\sigma_\theta = 0.389°$ (0.389–0.400° over steps). It is also insensitive to derivative errors: on a fixed mesh the bound is 0.394°, a 1.3 % change, and random errors of ± 0.3 % on every 300 nm-rib derivative spread it only between 0.364° and 0.420° (5th to 95th percentile).

An exhaustive search over all sets of up to eight structures that use one partial etch finds a lower bound with more rib widths and some TM strips. The best set of eight, TE strips of 400 nm and 600 nm, TM strips of 400 nm and 600 nm and ribs of 300, 400, 500 and 600 nm on a 90 nm slab, gives $\sigma_\theta = 0.139°$ (0.101–0.147° over steps). The best sets of four to seven structures give 0.435, 0.308, 0.153 and 0.145°. This set is more sensitive to derivative accuracy than the two-width set. With fixed-mesh derivatives for the 300 nm rib its bound becomes 0.118° (−15.3 %), the fixed-mesh and remeshed derivatives differing by at most 0.78 %, and random ± 0.3 % errors spread it between 0.109° and 0.189°. We therefore quote 0.12–0.14° for this set, with a derivative-accuracy uncertainty of about 15 %. Without the index unknowns (stage C, no priors) the best set of eight gives 0.124°.

A greedy search that adds one structure at a time reached the exhaustive optimum in 25 of the 32 stage, prior and slab cases we compared and fell short in 7, with a bound up to a factor of

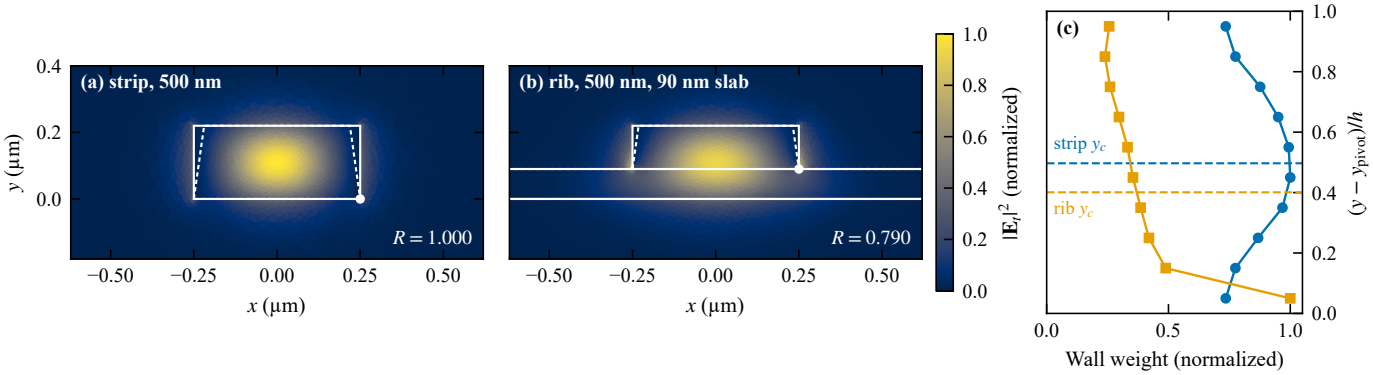


Fig. 1. Why a strip cannot see its sidewall angle. (a) Oxide-clad strip, 500 nm × 220 nm, and (b) rib, 500 nm wide on a 90 nm slab, both TE at $\lambda = 1.55\,\mu$m, with the transverse intensity $|\mathbf{E}_t|^2$, the nominal wall (solid) and a tilted wall (dashed; 8°, exaggerated for visibility) pivoting about the bottom corner (dot). The femwell angle/width ratio R is printed in each panel. (c) First-order wall-perturbation weight W along the right wall versus height above the pivot, normalized to its maximum, with the centroids y_c (dashed). The strip weight is symmetric about mid-height, so a tilt is exactly a width change ($R = 1$); the rib weight sits low on the wall, so $R < 1$.

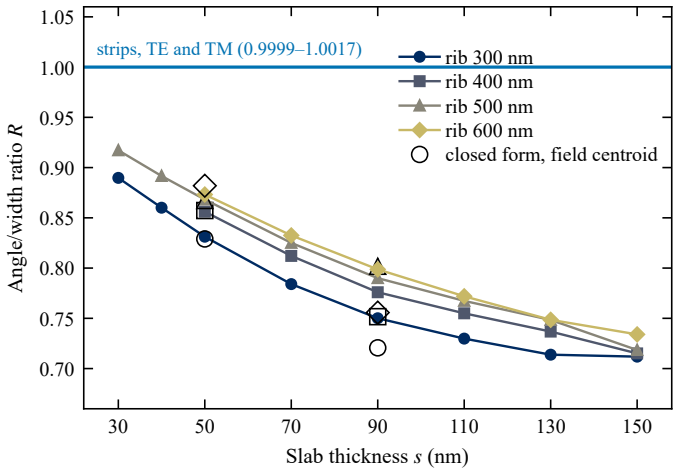


Fig. 2. Angle/width ratio R of the TE ribs versus slab thickness for four rib widths (femwell derivatives at step h; $\lambda = 1.55\,\mu$m, $t = 220$ nm, oxide-clad). Open symbols: closed form $R = (y_c - y_{\mathrm{pivot}})/(h/2)$ from the mode field at the 50 nm and 90 nm slabs. The line at $R = 1$ is the strip value, set by symmetry. Every rib breaks the strip degeneracy, and thinner slabs break it least.

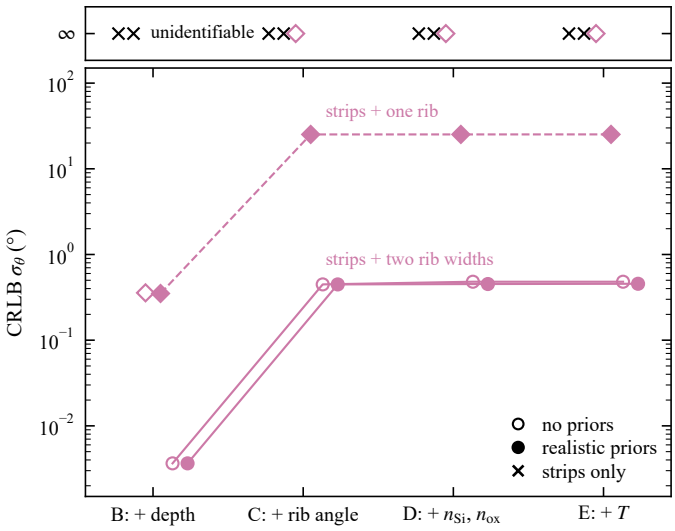


Fig. 3. Strip sidewall-angle CRLB as process unknowns are added (B: etch depth, rib angle tied to the strip angle; C: separate rib angle; D: silicon and oxide indices; E: temperature), for strips alone (crosses), strips plus one 500 nm rib (diamonds) and strips plus ribs of 500 nm and 300 nm (circles), all on a 70 nm slab, without priors (open) and with realistic priors (filled), at $\sigma_\lambda = 1$ pm. Cases with a singular Fisher matrix are drawn in the band marked ∞. A second width of the same rib is what keeps the angle identifiable once the partial etch has its own angle.

1.12 larger. At the best-case 0.1 pm wavelength precision both bounds shrink tenfold, to 0.0408° for the two-width set and 0.0139° for the best set; they scale linearly with σ_λ (0.196° and 0.775° for the two-width set at 0.5 and 2 pm).

E. Monte Carlo fits reach the bound

Fig. 4 compares the nonlinear fits with the bound. For the two-width set the fitted angles have a standard deviation of 0.398° (0.385–0.410°, 95% bootstrap interval), 1.02 times the CRLB, with a bias of $(-0.0150 \pm 0.0089)°$; none of the 2000 fits failed, ended ambiguous or reached the box edge. For the best set the spread is 0.132° (0.125–0.140°), 0.952 times the CRLB, with a bias of $(0.0075 \pm 0.0030)°$; 4 fits ended at the box edge and none failed. The spread is slightly below the femwell CRLB because the Monte Carlo uses the surrogate model, whose own bound at nominal is 0.1249°: its 300 nm-rib slopes are a smooth fit through the remeshing scatter. Measured against that bound the fits are slightly less efficient than the CRLB, as expected for a nonlinear estimator. With one rib the fits find the valley predicted by the symmetry argument. The linearized bound is 29.52°, so the fit box of ±3° confines the estimates rather than the data: 1190 of the 2000 fits end at the edge of the box, 12 are ambiguous and 20 fail, and the fitted widths and angles lie on a line of slope 3.842 nm per degree with a correlation of 0.999998, against $t\tan(1°) = 3.8401$ nm (Fig. 4(a)).

F. Slab thickness

For the two-width set the bound depends strongly on the slab (Fig. 5). It is lowest at 50 nm (0.389°) and rises slowly toward thinner slabs, to 0.403° at 40 nm and 0.414° at 30 nm, and

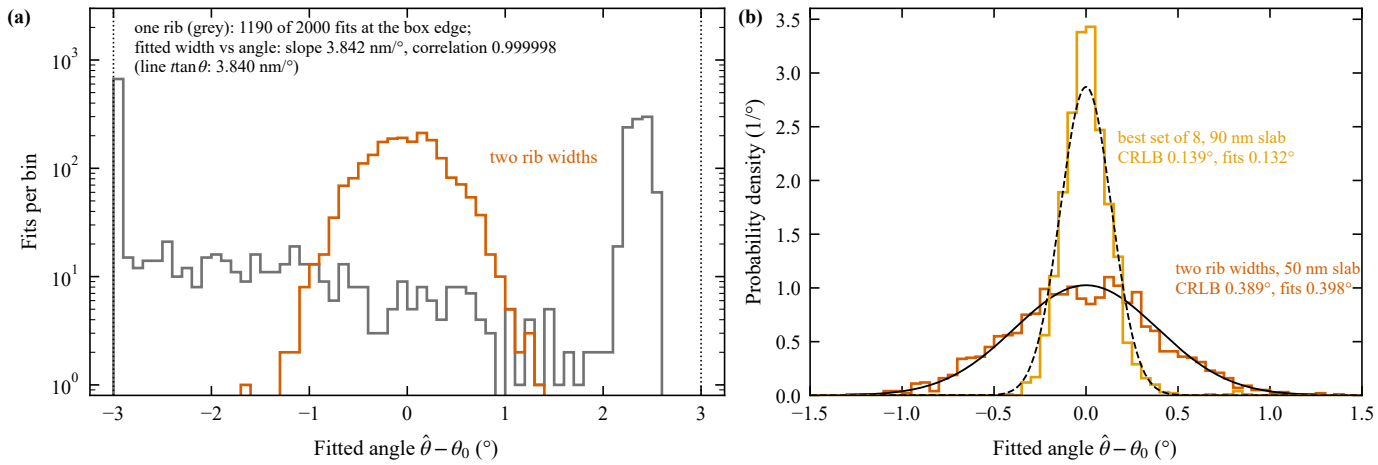


Fig. 4. Monte Carlo fits of 2000 simulated noisy measurement sets per set at $\sigma_\lambda = 1\,\mathrm{pm}$ (stage D, realistic priors, bounded nonlinear least squares). (a) Histograms of the fitted strip angle for strip TE rings at five widths plus one 500 nm rib (grey) and plus ribs of 300 nm and 500 nm (orange), both on a 50 nm slab; dotted lines mark the $\pm 3°$ fit box. With one rib the fits spread over the whole box along the width/angle valley. (b) Fitted-angle histograms for the two-width set and the best set of eight (90 nm slab) with Gaussians of the CRLB width (lines): with two or more rib widths the fit scatter matches the bound.

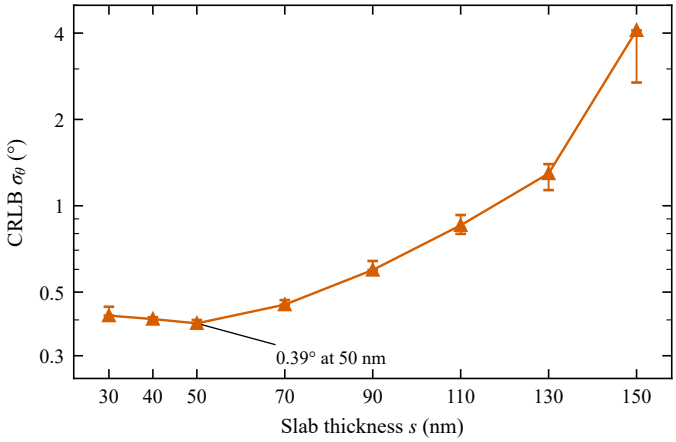


Fig. 5. CRLB of the two-width set (strip TE rings at 400–600 nm plus ribs of 300 nm and 500 nm) versus slab thickness, stage D, realistic priors, $\sigma_\lambda = 1\,\mathrm{pm}$, $\lambda = 1.55\,\mu\mathrm{m}$. Bars span the derivative steps $h/2$, h, $2h$. The bound is lowest at 50 nm and grows tenfold by 150 nm.

quickly toward thicker ones, to 0.597° at 90 nm and 4.089° at 150 nm. Among the slabs simulated the optimum is therefore 50 nm, not the thinnest slab. Thinner slabs bring R closer to 1 (Fig. 2), but we have not separated that effect from the other ways the slab changes the two rib modes.

G. Cost of ignoring the angle

When strip rings are fitted with the angle held at its nominal value, a real 1° tilt is absorbed as a width change of $-3.84\,\mathrm{nm}$. Using that width to predict a 500 nm rib ring on a 90 nm slab, whose angle sensitivity differs ($R < 1$), mispredicts its resonance by $\delta\lambda = \lambda\,\delta n_{\mathrm{eff}}/n_{\mathrm{g}} = -711\,\mathrm{pm}$ per degree of unmodeled tilt.

VI. Discussion

For a foundry or a test-chip designer, the practical message is about which structures to place. Width and thickness monitors made of strips, however many and in whatever polarization, cannot report the sidewall angle of an oxide-clad full etch. Adding one rib from the partial etch does not help when that etch is a separate step with its own depth and angle, and better depth metrology does not change this. Adding two or more widths of the same rib does: seven structures, five strips and two ribs on a 50 nm slab, each read by a ring and an MZI, bound the strip angle at 0.39° for a 1 pm wavelength precision, and the bound is reached by an ordinary nonlinear fit. These structures occupy little area and use the same readout as existing ring monitors, so they could sit next to them in every reticle field and report the angle where SEM or AFM cross-sections are not taken.

We see this as a complement to dimensional metrology rather than a replacement. Cross-sections and AFM measure the profile directly at a few sites and remain the reference for calibrating any optical extraction; scatterometry is fast but model-based [2], [3]. The optical structures measure the quantity that matters to the circuit, the effect of the angle on the guided modes, over the whole wafer, and they need a calibrated model of the cross-section to convert that effect into degrees.

Several limitations bound the conclusions. The results are simulations; no device was fabricated or measured. The symmetry argument holds for oxide above and below the core. An air-clad strip is not mirror-symmetric, and its angle/width ratio in our solves is 0.9782 rather than 1, but we computed no bound for air-clad sets and make no claim about them. The walls are ideal trapezoids tilting about their bottom edge, without corner rounding or roughness; the material indices are nominal values without dispersion; and n_{g} uses a fixed 5 nm wavelength step. None of these choices was varied. All structures in a set are assumed to share the same width offset, thickness and angles, as if placed close enough together that variation between them is negligible; local variation within a set was not modeled. Rings are assumed to have known resonance orders and lengths, so that each resonance gives n_{eff} directly.

The noise model starts from a cited wavelength precision, but the step to σ_{n_g} and σ_ϕ is ours and assumes an MZI fringe is located as precisely as a ring resonance. The bounds scale linearly with σ_λ and depend on the other assumptions by tens of percent: halving or doubling the rib noise relative to the strips changes the two-width bound by $-47.9\,\%$ and $97.5\,\%$, and the ring length used in the conversion changes it by $-49.4\,\%$ at $50\,\mu\mathrm{m}$ and $97.8\,\%$ at $200\,\mu\mathrm{m}$. The priors matter much less, at most $0.6\,\%$ for factor-of-two changes. Even the two-width bound carries a numerical uncertainty of about $5\,\%$, the change when the domain is enlarged. The best eight-structure set is limited by derivative accuracy: its bound moves by $-15.3\,\%$ between remeshed and fixed-mesh derivatives and by $-27.5\,\%$ at half the derivative step, which is why we quote it as a range. Whether the rib and strip angles are independent, partly linked or equal depends on the process; we treat independence as the default because the two etches are separate steps, and show in Section V how much a link would help. Finally, the analysis covers TE ribs only and the $1.55\,\mu\mathrm{m}$ band.

The next step is a test chip that carries the two-width set and the best set next to conventional width monitors, measured at controlled temperature and compared with SEM and AFM cross-sections of the same dies. On the modeling side, TM ribs, air cladding and other platforms such as silicon nitride and thin-film lithium niobate, where the cladding and the etch profile differ, are open.

VII. Conclusion

Oxide-clad strip waveguides cannot report their sidewall angle, because their mirror symmetry makes a tilt exactly equivalent to a width change to first order. A partially etched rib breaks the symmetry, but when it comes from a separate etch a single rib width brings as many unknowns as observables and the strip angle stays unidentifiable, with or without a precise etch depth. Two widths of the same rib recover it: simulations indicate a bound of 0.39° for five strips and two ribs on a $50\,\mathrm{nm}$ slab at a $1\,\mathrm{pm}$ resonance-wavelength precision, and about 0.12–0.14° for the best set of eight, with nonlinear fits reaching the bound. A fabricated test chip with cross-section ground truth is needed to confirm these numbers.

Acknowledgment

acknowledgments: funding, lab, people

References

[1] Z. Lu, J. Jhoja, J. Klein, X. Wang, A. Liu, J. Flueckiger, J. Pond, and L. Chrostowski, "Performance prediction for silicon photonics integrated circuits with layout-dependent correlated manufacturing variability," *Opt. Express*, vol. 25, no. 9, p. 9712, 2017.

[2] N. G. Orji, M. Badaroglu, B. M. Barnes, C. Beitia, B. D. Bunday, U. Celano, R. J. Kline, M. Neisser, Y. Obeng, and A. E. Vladar, "Metrology for the next generation of semiconductor devices," *Nat. Electron.*, vol. 1, no. 10, pp. 532–547, 2018.

[3] C. Raymond, "Overview of scatterometry applications in high volume silicon manufacturing," in *AIP Conf. Proc.*, vol. 788, 2005, pp. 394–402.

[4] Y. Xing, J. Dong, S. Dwivedi, U. Khan, and W. Bogaerts, "Accurate extraction of fabricated geometry using optical measurement," *Photon. Res.*, vol. 6, no. 11, p. 1008, 2018.

[5] S. Dwivedi, A. Ruocco, M. Vanslembrouck, T. Spuesens, P. Bienstman, P. Dumon, T. Van Vaerenbergh, and W. Bogaerts, "Experimental extraction of effective refractive index and thermo-optic coefficients of silicon-on-insulator waveguides using interferometers," *J. Lightw. Technol.*, vol. 33, no. 21, pp. 4471–4477, 2015.

[6] S. Janz, S. Dedyulin, D.-X. Xu, M. Vachon, S. Wang, R. Cheriton, and J. Weber, "Measurement accuracy in silicon photonic ring resonator thermometers: Identifying and mitigating intrinsic impairments," *arXiv preprint arXiv:2306.15643*, 2023.

[7] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Upper Saddle River, NJ: Prentice Hall, 1993.

[8] B. J. Frey, D. B. Leviton, and T. J. Madison, "Temperature-dependent refractive index of silicon and germanium," in *Proc. SPIE 6273*, 2006, arXiv:physics/0606168.

[9] D. B. Leviton and B. J. Frey, "Temperature-dependent absolute refractive index measurements of synthetic fused silica," *arXiv preprint arXiv:0805.0091*, 2008.

[10] S. G. Johnson, M. Ibanescu, M. A. Skorobogatiy, O. Weisberg, J. D. Joannopoulos, and Y. Fink, "Perturbation theory for Maxwell's equations with shifting material boundaries," *Phys. Rev. E*, vol. 65, no. 6, p. 066611, 2002.

[11] H. Gehring, "femwell: mode solver for photonic and electric waveguides based on FEM, version 0.1.12," https://github.com/HelgeGehring/femwell, documentation https://HelgeGehring.github.io/femwell/.

[12] T. Gustafsson and G. McBain, "scikit-fem: A Python package for finite element assembly," *J. Open Source Softw.*, vol. 5, no. 52, p. 2369, 2020.

[13] C. Geuzaine and J.-F. Remacle, "Gmsh: A 3-D finite element mesh generator with built-in pre- and post-processing facilities," *Int. J. Numer. Methods Eng.*, vol. 79, no. 11, pp. 1309–1331, 2009.

[14] "EMpy (ElectroMagneticPython), version 2.2.3," https://github.com/lbolla/EMpy.

[15] L. H. Jiao and F. Pikus, "Silicon waveguides and temperature-dependent performance," in *Mastering Optoelectronics – Fundamentals, Applications, and Innovations [Working Title]*. IntechOpen, 2025.